



# Michael Havrylyuk

mhavry@uw.edu | [Github](#) | [Linkedin](#)

Additional Photos and Videos are hosted on my Immich Server; photos will be included in this document where possible.  
[Immich Server Link](#)

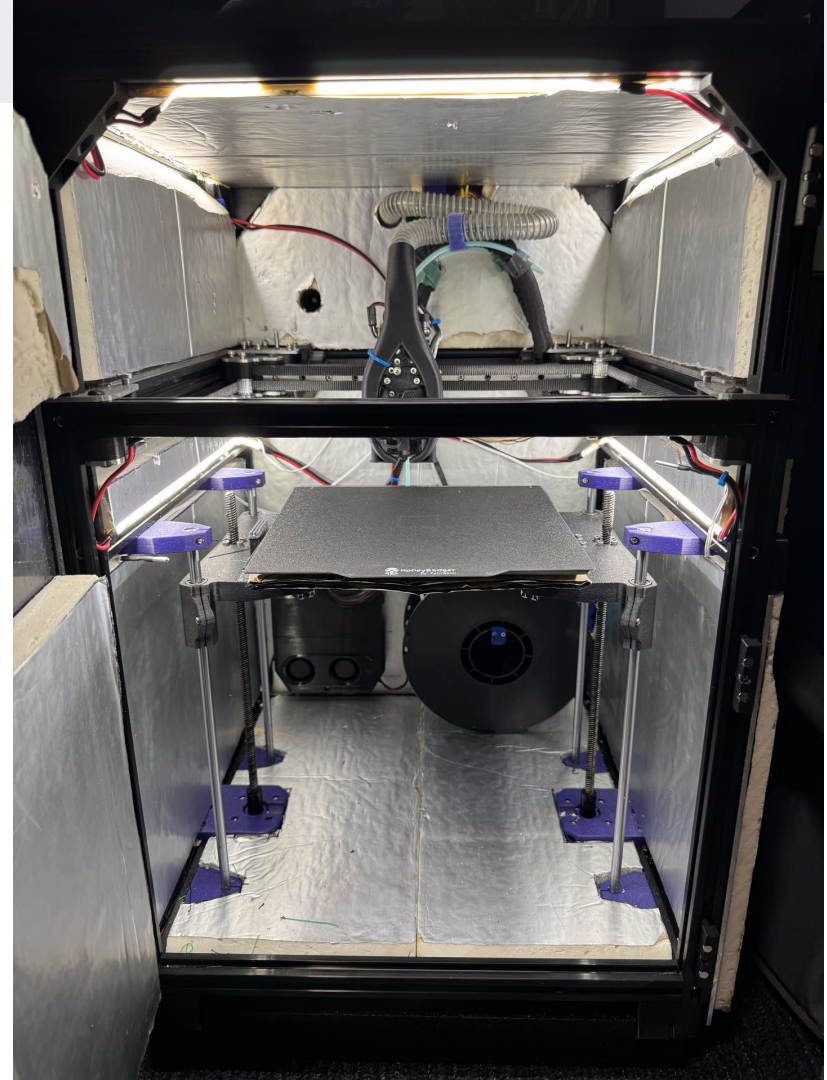


# 3D Printer Build

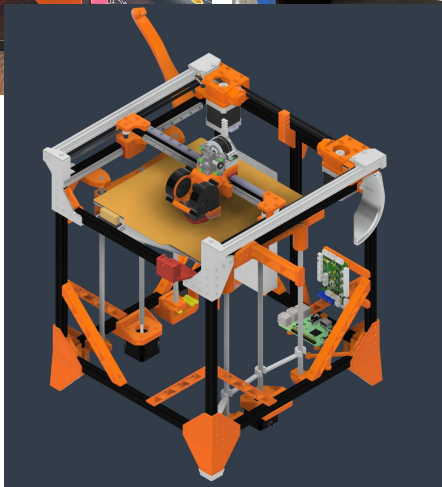
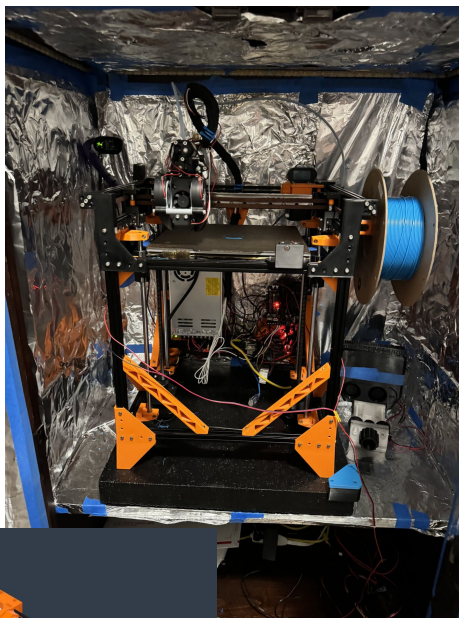
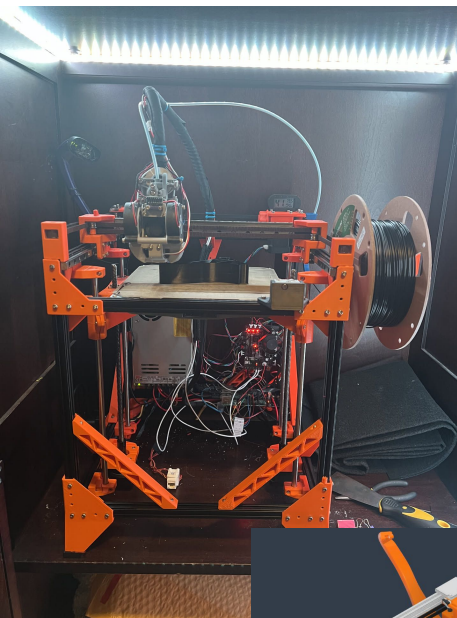
Hobby / Project

I have spent the last several years iterating and rebuilding my custom-built 3D printer into a high-temp machine. I am now able to print ABS and ASA at 70 °C chamber temperature, which allows my parts to have superior layer adhesion and strength.

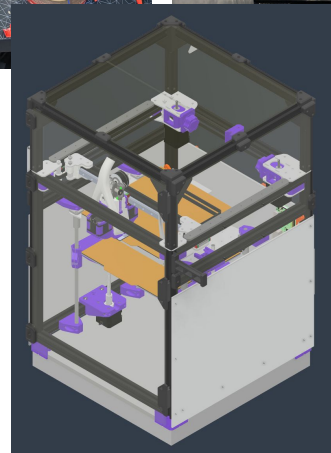
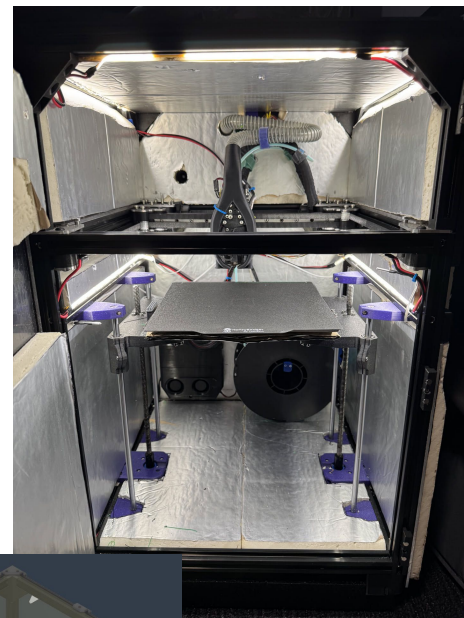
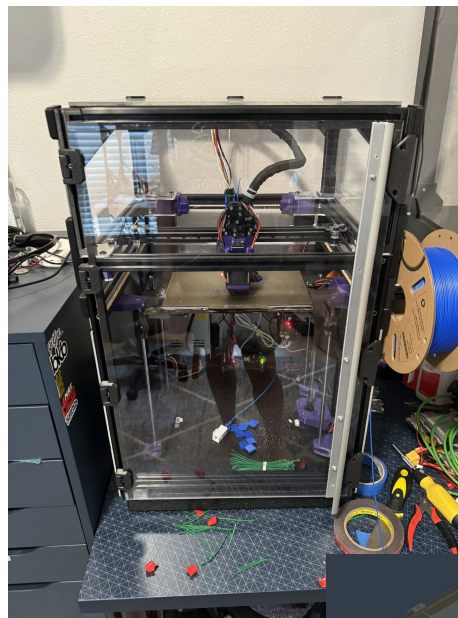
Skills Used: Soldering, Fusion 360 CAD, Fusion 360 FEA, ANSYS CFD, SimScale CFD, Klipper Configuration, Thermodynamics, Resonance Analysis



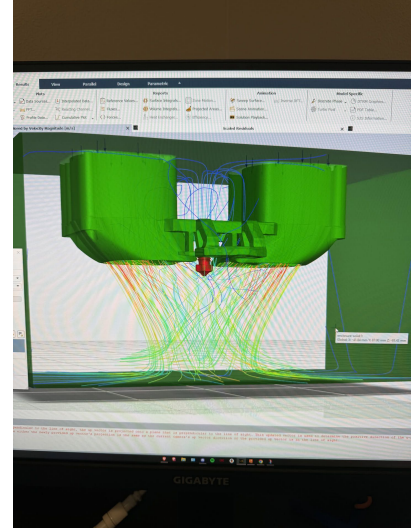
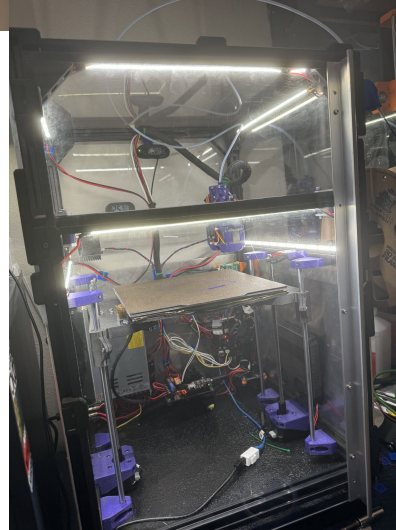
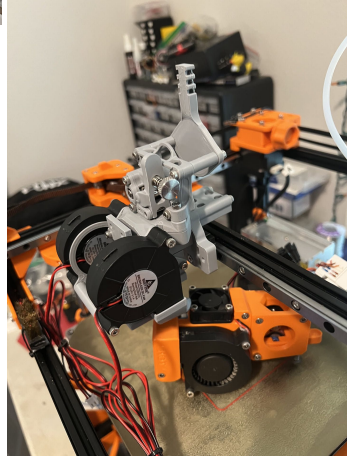
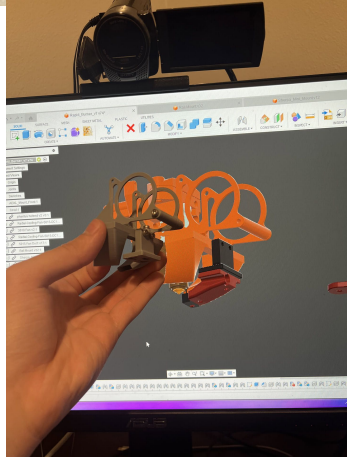
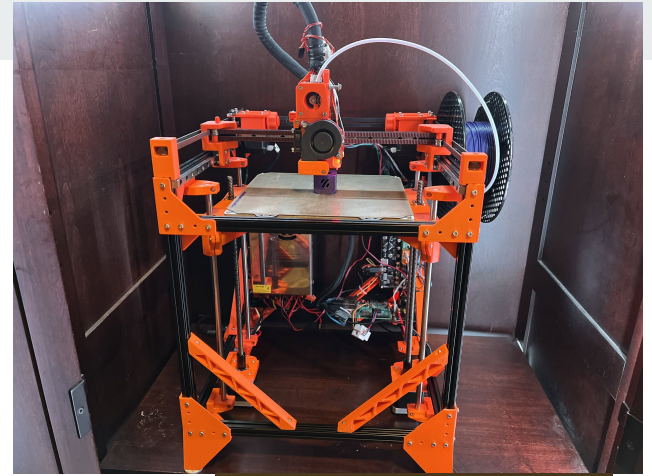
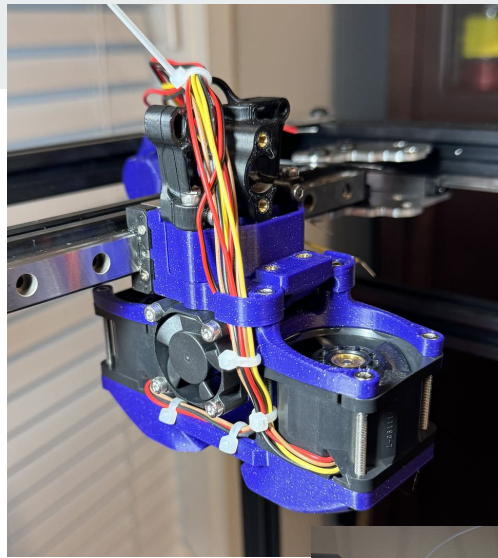
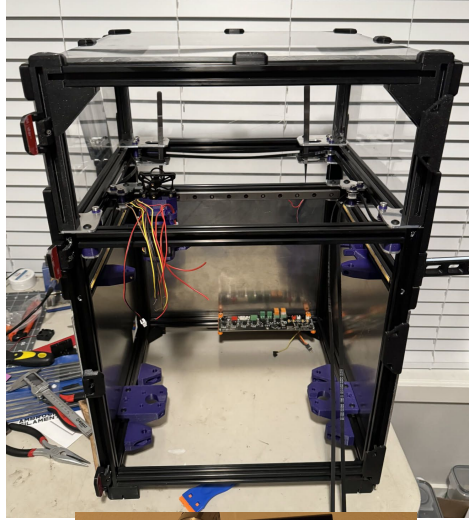
V1 Printer



V2 Printer



# More Photos



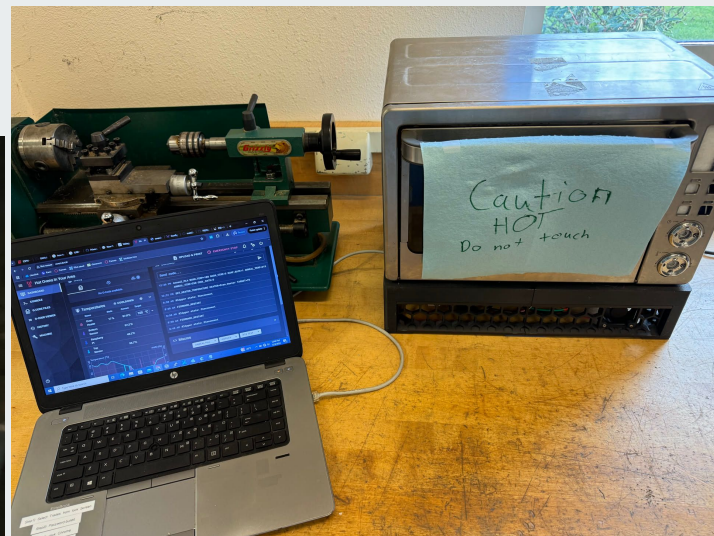
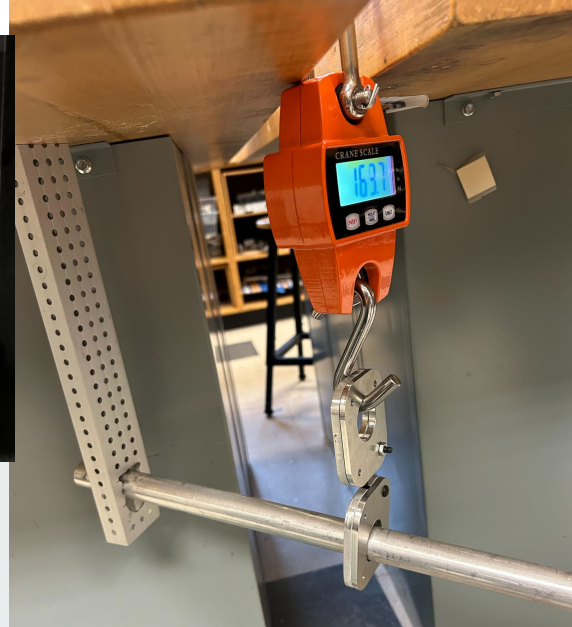
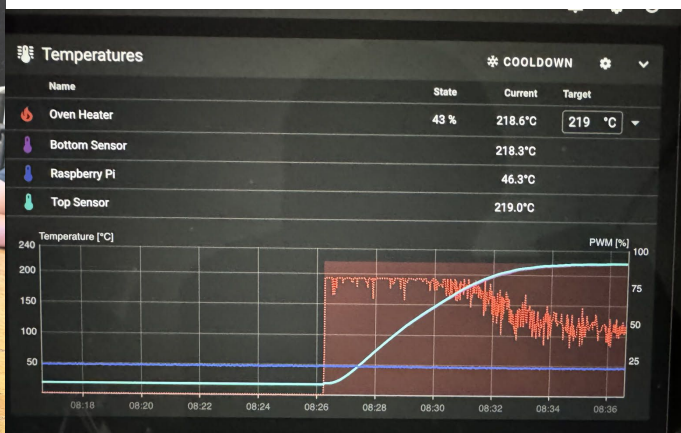
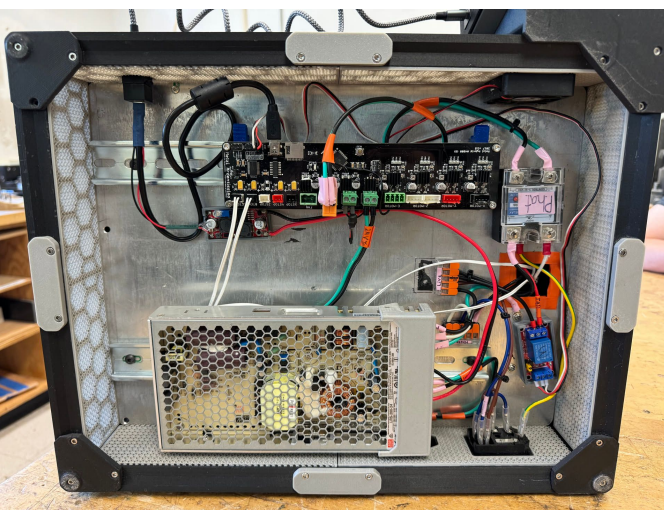
# PID Oven

Problem: Annealing is super complex, create a device that could conceivably be a product with not much adaptation and use it to create annealing profiles to optimize for strength and heat deflection temperature

Process: Bought a convection oven off of facebook marketplace. Installed 3d printer electronics into it. Installed Klipper and did tuning. Created intuitive user interface using KlipperScreen

Result: Characterized PPS-CF and PET-CF filament at different annealing temperatures and discovered Sulfur dioxide deposits in high temperature annealed PPS-CF samples.

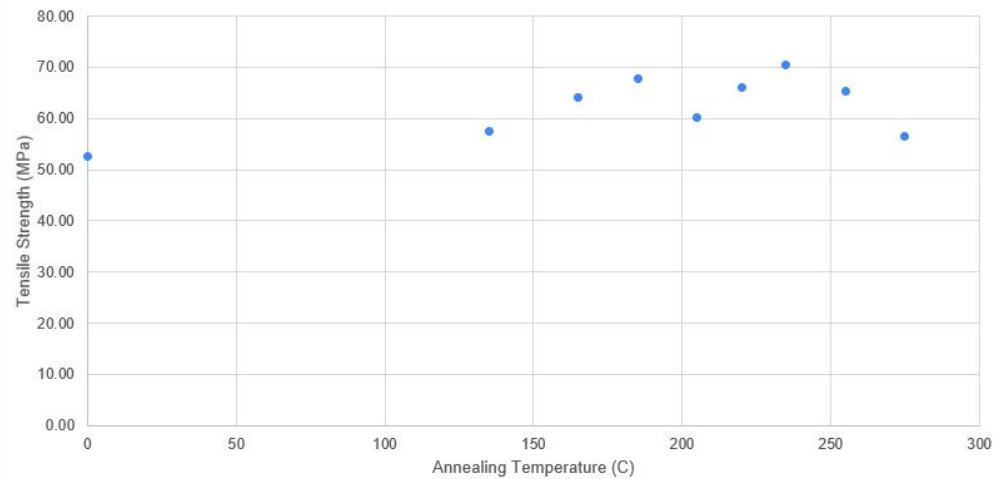




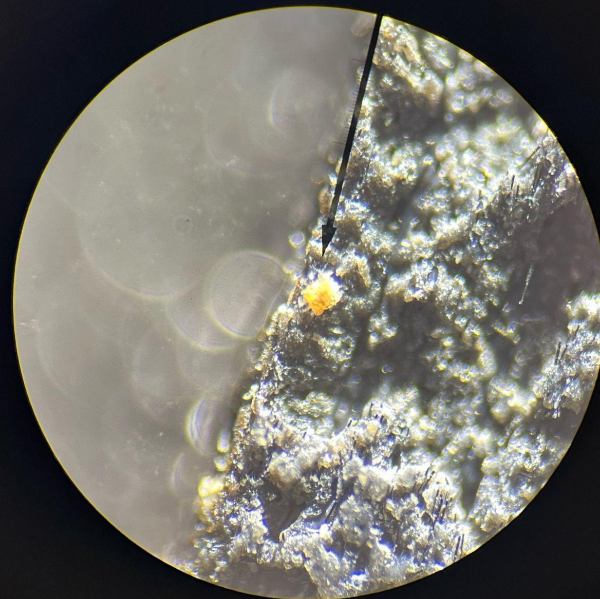
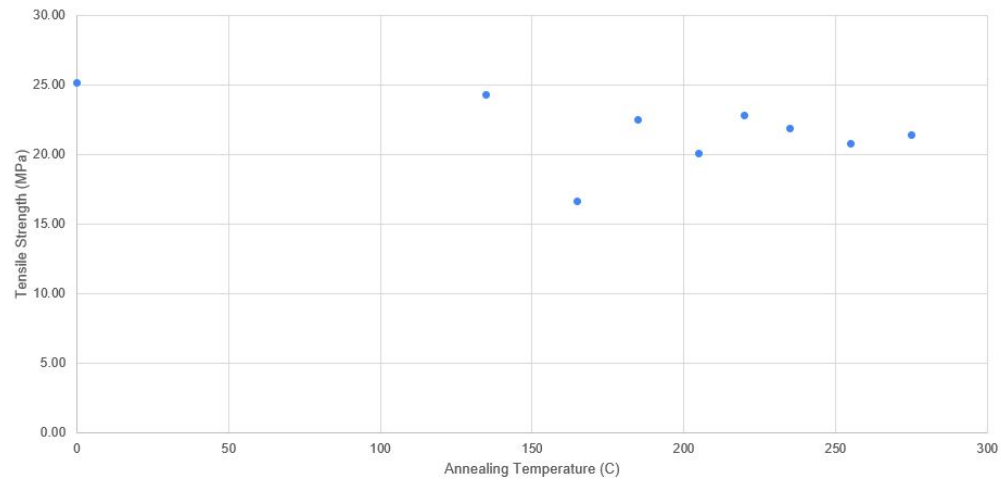
| A                      | O   | P    |
|------------------------|-----|------|
| PET-CF (Unannealed) E  | <70 | 75   |
| PET-CF (100c Anneal) 3 | 100 | 146  |
| PET-CF (110c Anneal) J | 105 | 146  |
| PET-CF (120c Anneal) 7 | 105 | 150  |
| PET-CF (130c Anneal) 8 | 110 | 165  |
| PET-CF (140c Anneal) H | 125 | 182  |
| PET-CF (150c Anneal) E | 127 | 185  |
| PET-CF (160c Anneal) V | 130 | 187  |
| PET-CF (170c Anneal) J | 138 | 203  |
| PET-CF (180c Anneal) F | 145 | 208  |
| PET-CF (190c Anneal) 8 | 160 | >216 |

Sheet1

PPS-CF XY Annealing Tensile Strength



PPS-CF Z Annealing Tensile Strength



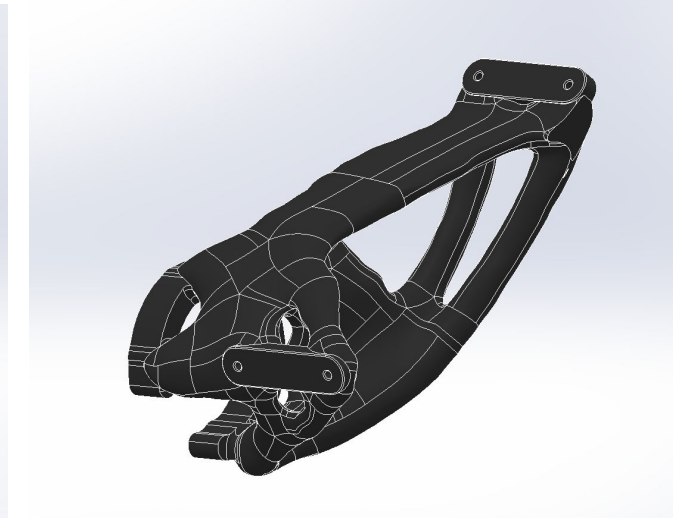
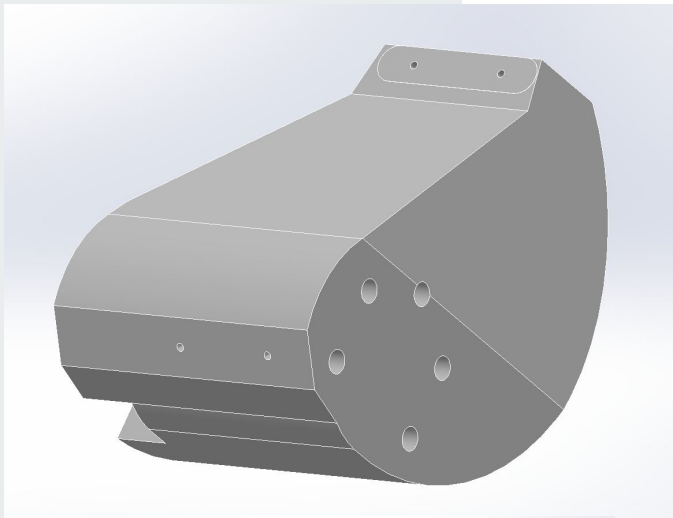


# Robotics

Advanced Robotics at the University of Washington - Engineer Squad



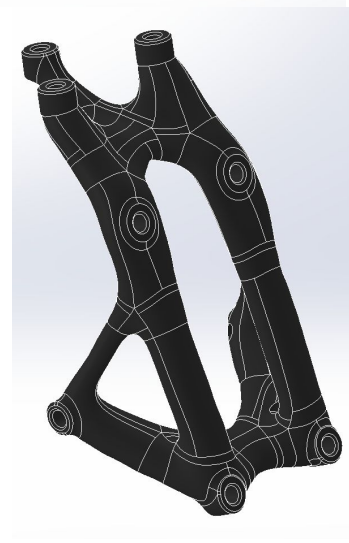
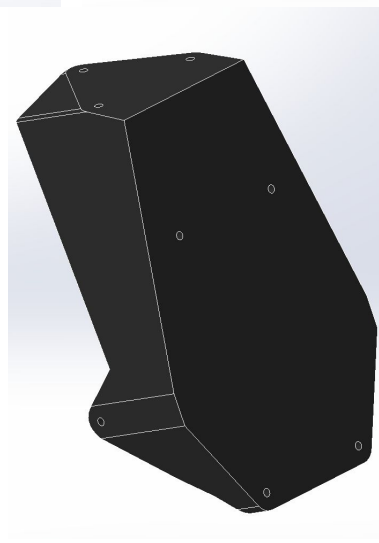
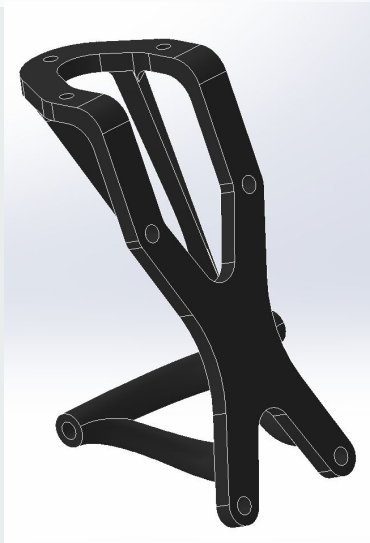
# nTop



Problem: Weight is at a premium, we need to save weight where ever we can.

Process: Design rough geometry in Solidworks, create nTop Topology optimization notebook, run topology optimization, export and clean up in Solidworks

Result: Up to 11x increase in stiffness to weight ratio





# Robotics

FRC Team 2811 - Stormbots - Mechanical Design and Build Lead

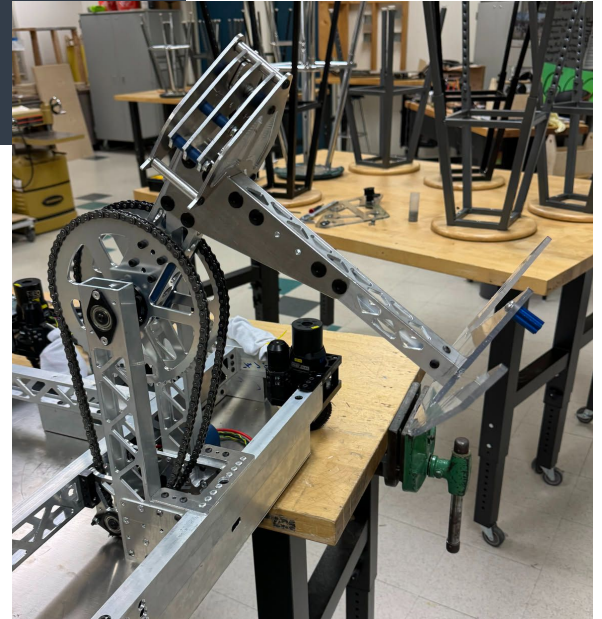
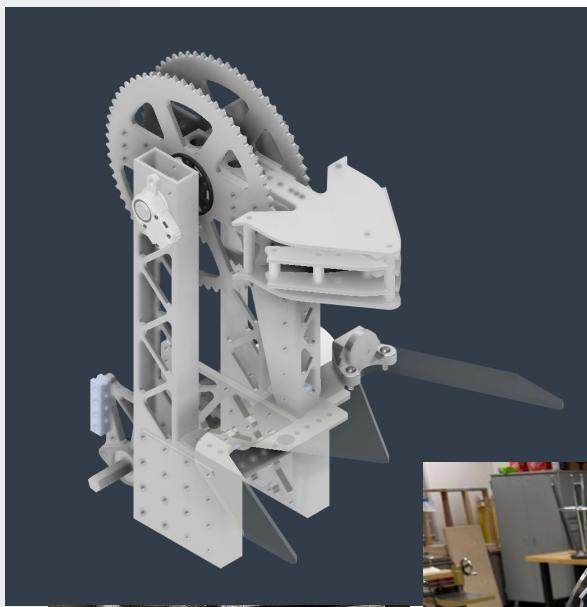


# Reefscape Climber

Problem: Climb on a Metal Cage while only holding on to the poles or bottom

Process: Prototyped with wood and metal. Designed in Fusion 360. 5 iterations

Result: Consistent and easy climbing



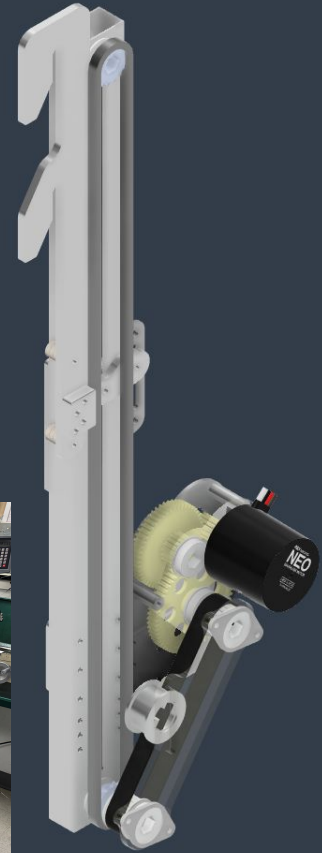
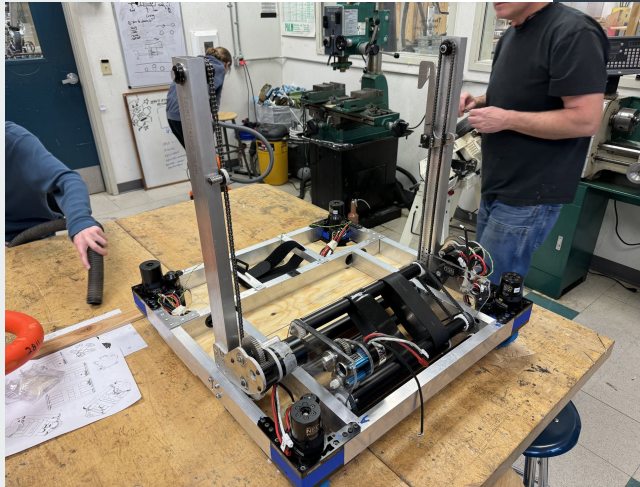
---

# Crescendo Climber

Problem: Climb as high as possible in order to allow scoring in a high goal while starting below 24 inches, chain to climb on is at 27 inches

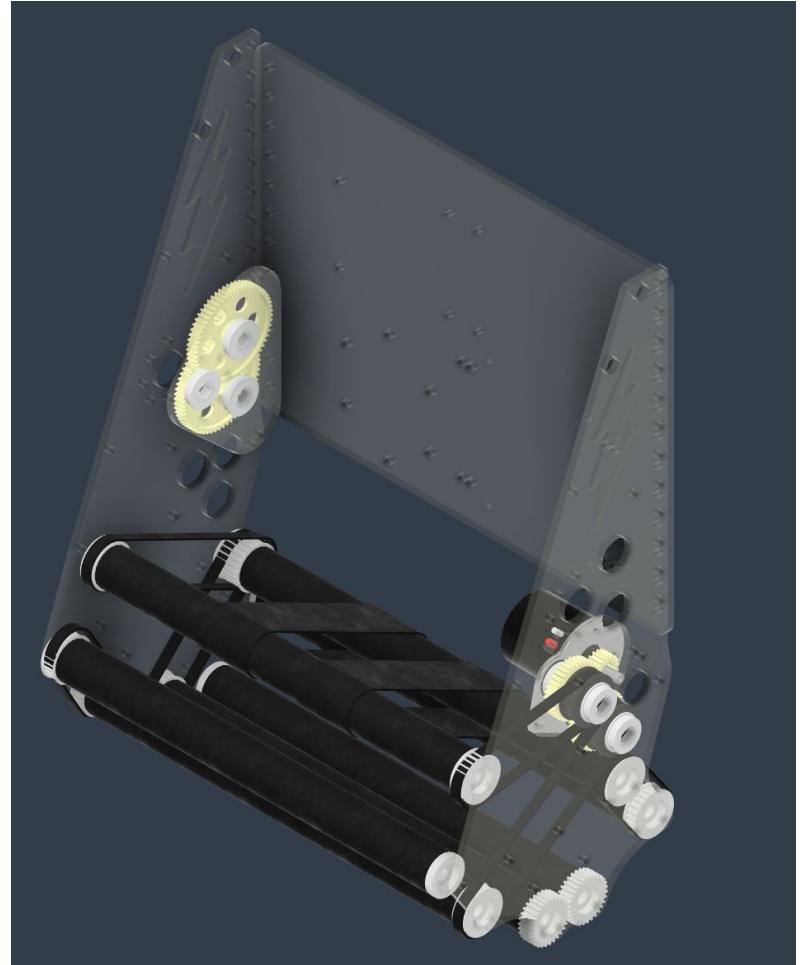
Process: Brainstorm possible options, design in Fusion 360. 3 iterations

Result: Ranked 17th out of 3477 in the world on climbing performance



# Crescendo Intake

Problem: Pickup foam rings off of ground from both directions  
Process: Prototype with different wheels and rollers. Design in Fusion 360. 4 iterations  
Result: No jams and high throughput capability. Mechanism for scoring while climbing mounted on the intake.



# Crescendo

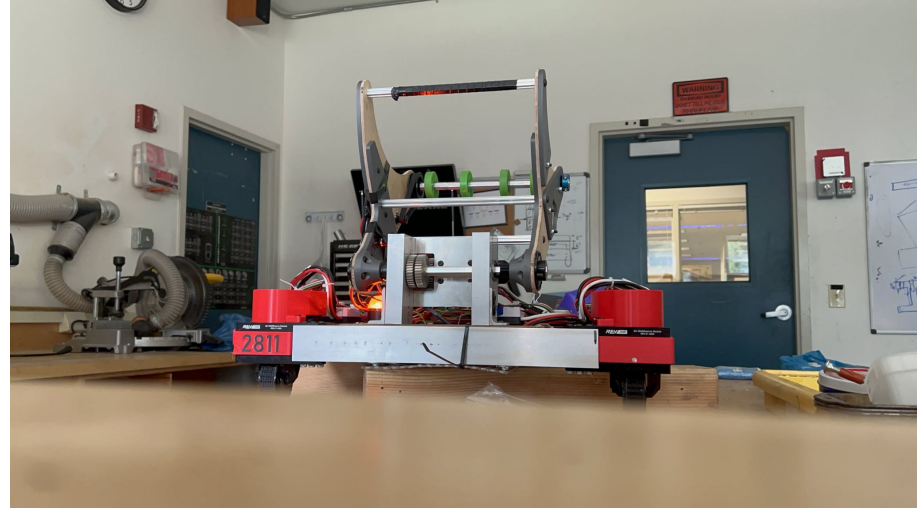
Both subsystems were very reliable.





# Offseason Swerve Base

I was also the lead designer, builder, and driver for Stormbots' first-ever Swerve Drive Robot during the offseason. This robot was designed for the Charged Up competition and to be built as cheaply and quickly as possible. It won its first and only event.



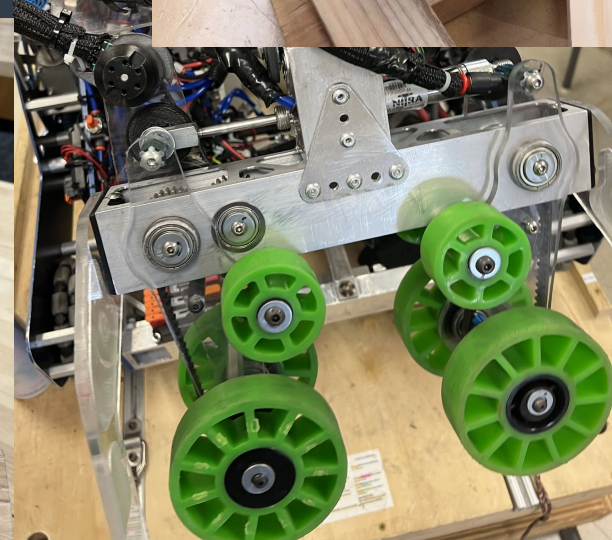
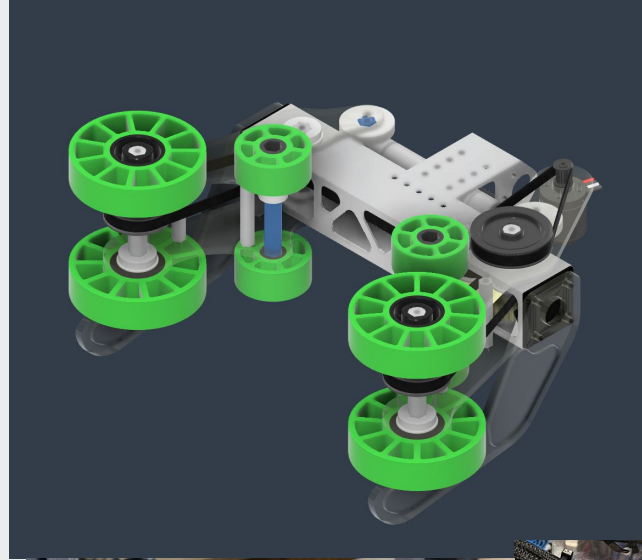


# Charged Up Roller Claw

Problem: Pick up cones and inflatable cubes from the ground and a shelf.

Process: Brainstorm and do research. Prototype with wood. Design using Fusion 360. 2 iterations

Result: Mounts to a pivoting wrist joint. Pneumatic actuation to open and close the claw. Reliable cone and cube acquisition.

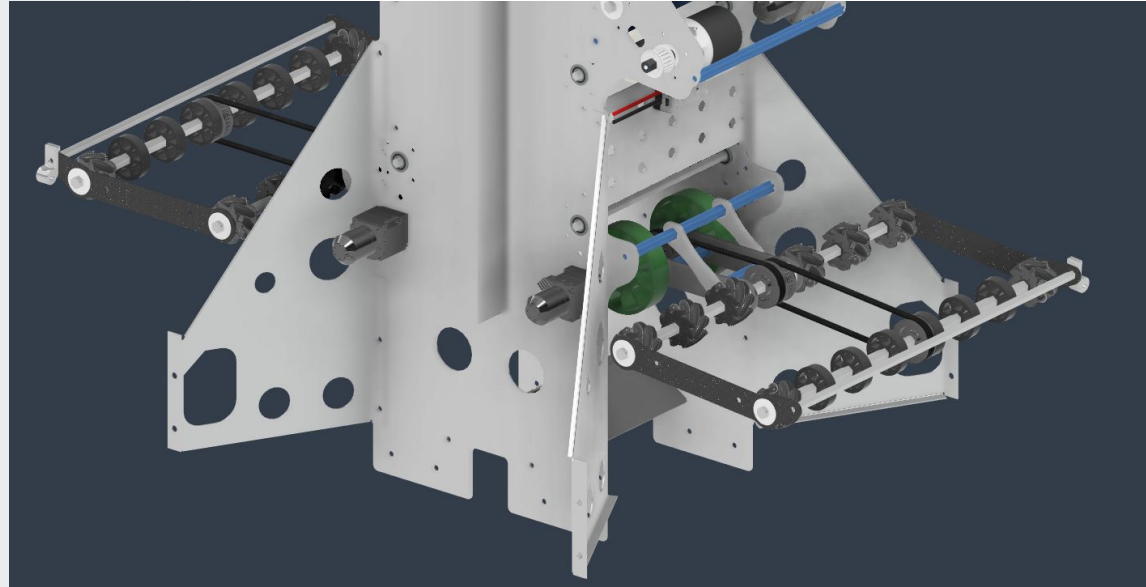


# Rapid React Passthrough

Problem: Transfer 24cm tennis balls from 2 different sources and stage them to be shot out of a shooter. Be very light.

Process: Prototype and do research. Design in Fusion 360.  
2 iterations

Result: Reliable acquisition of balls from both sides.





# Robotics

Offseason FRC Team 9999 - SK Robotics - Mechanical Design

# Elevator

Problem: Place PVC Pipe (Coral) at 4 scoring locations quickly and precisely.

Process: Research. Design in Fusion 360.

Result: Dual motor drive with internally routed HTD5 belt for power transmission. Continuously rigged belt.

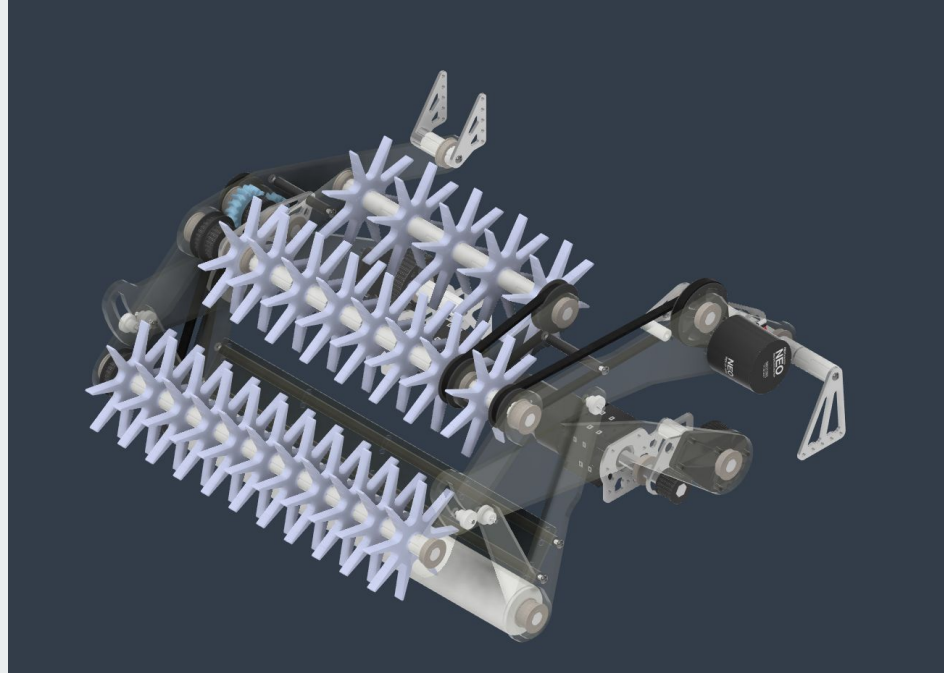


# Intake

Problem: Pick up PVC Pipe (coral) in any orientation on the ground

Process: Research. Design in Fusion 360.

Result: Fast acquisition of coral from any orientation. Can fold into the robot if it collides with something.

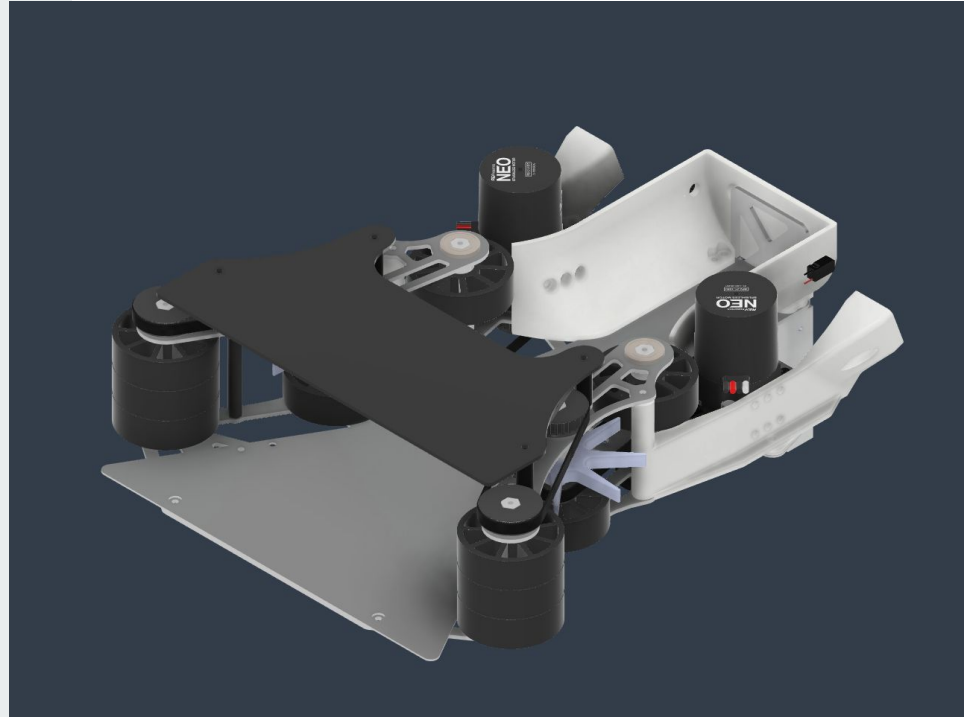


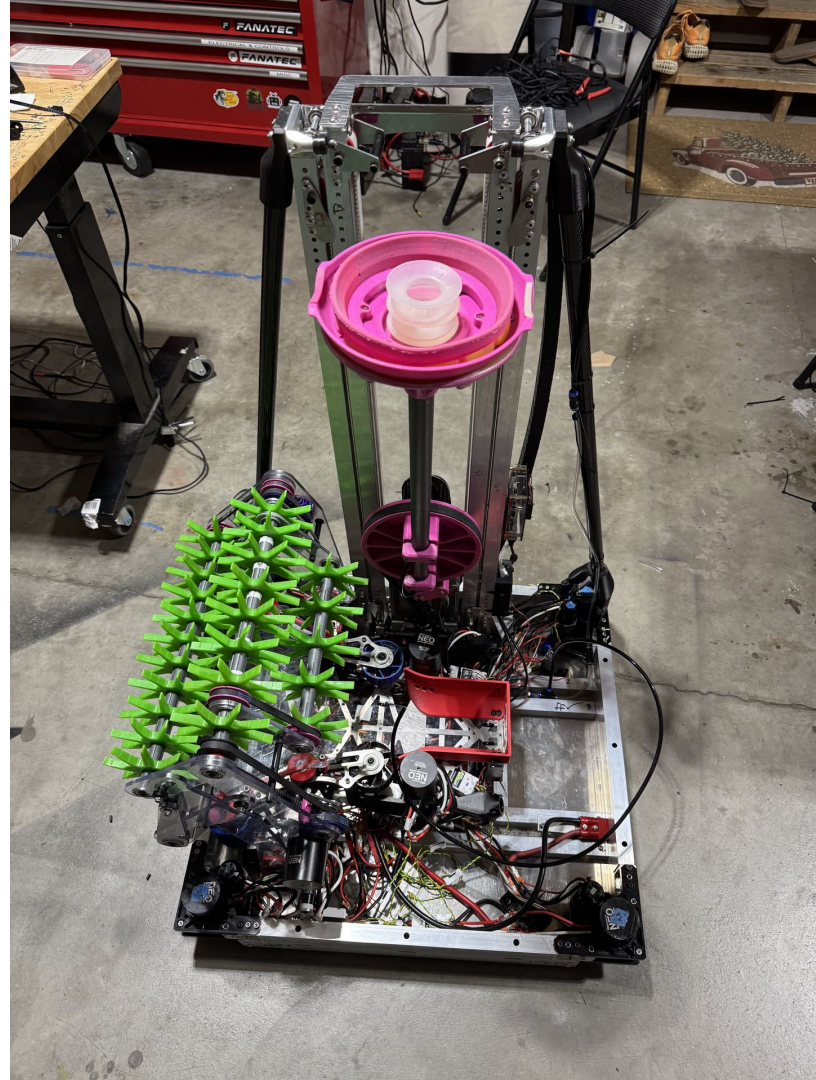
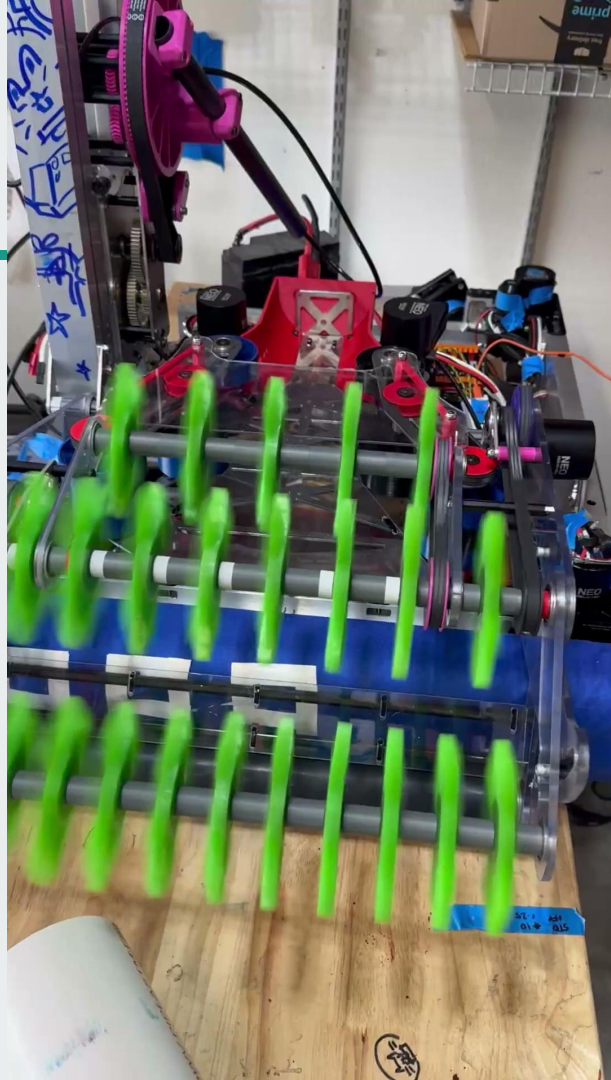
# Passthrough

Problem: Reorient PVC Pipe (coral) from any orientation into a known position.

Process: Research. Design in Fusion 360.

Result: Each side being driven individually means coral jams can be detected and fixed. 3D printed coral carrier ensures coral is in the exact same position every time.





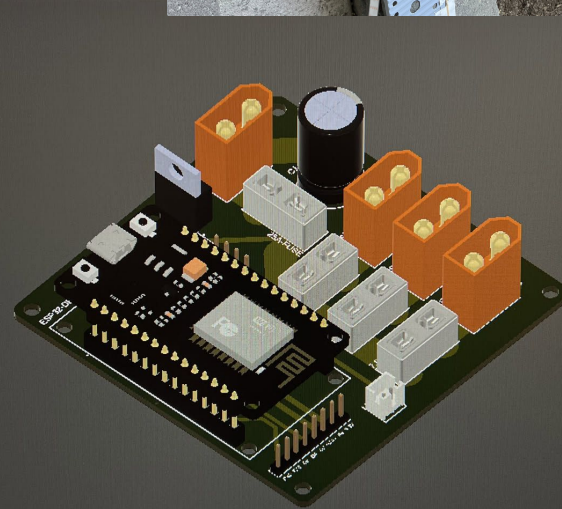


# Retaining Wall LEDS

Project



Skills Used: Soldering, Fusion 360  
PCB, Waterproofing, Designing for  
scalability, ESPHome software





# Electric Longboard

Project



Skills Used: 3D Printing, Inventor, Spot Welding, Programming, Soldering, VESC Software Configuration

